

RESEARCH ARTICLE OPEN ACCESS

Identification of Maternal Health Risk From Optimal Features Using Explainable Machine Learning

Mohammad Mamun^{1,2} | Safiul Haque Chowdhury¹  | Md. Omar Faruq³ | Mir Mohammad Azad⁴ | Bishakha Rani Biswas⁵ | Mohammed Ibrahim Hussain² | Muhammad Minoar Hossain²

¹Department of Computer Science and Engineering, Jahangirnagar University, Dhaka, Bangladesh | ²Department of Computer Science and Engineering, Bangladesh University, Dhaka, Bangladesh | ³Department of Computer Science and Engineering, Bangladesh Army University of Engineering & Technology, Natore, Bangladesh | ⁴Department of Computer Science and Engineering, World University of Bangladesh, Dhaka, Bangladesh | ⁵Maine College of Engineering and Computing, The University of Maine, Orono, Maine, USA

Correspondence: Mohammad Mamun (mamun.faster@gmail.com)

Received: 7 September 2025 | **Revised:** 15 October 2025 | **Accepted:** 30 October 2025

Funding: The authors received no specific funding for this work.

Keywords: ensemble model | explainable artificial intelligence | feature optimization | machine learning

ABSTRACT

Maternal health is a significant global crisis that affects women every day, resulting in severe complications related to pregnancy or childbirth. As a response to this urgent need for effective risk management, we have developed an automated system for maternal health risk (MHR) prediction that utilizes machine learning (ML) and explainable artificial intelligence (XAI). Our research aims to enhance the accuracy and efficiency of risk assessment by employing rigorous preprocessing techniques on a dataset of 1014 samples. To achieve this goal, we employ 10 ensemble ML models and use diverse feature optimization methods such as principal component analysis (PCA), linear discriminant analysis (LDA), and recursive feature elimination (RFE). To clearly understand the decision-making processes of the selected ensemble model, we employ XAI as Shapley additive explanations (SHAP) plots, local interpretable model-agnostic explanation (LIME) plots, and individual conditional expectation (ICE) plots. Our evaluation includes various ML performance metrics and a variety of statistical measures, which demonstrate that the Bagging, in conjunction with PCA, emerges as the optimal model, achieving an impressive accuracy of 99.51%. This research uses 10-fold cross-validation throughout the whole analysis. By emphasizing proactive risk detection and personalized interventions, we aim to improve understanding of MHR and help women worldwide make informed decisions.

1 | Introduction

Maternal health risk (MHR) includes factors that can lead to unfavorable outcomes before and after childbirth and is closely linked to maternal mortality. It is critical to identify and handle this risk to prevent complications that may result in maternal death. Premature mortality emphasizes taking proactive steps to reduce risks during pregnancy. In contrast, postpartum concerns

recognize that risks to a mother's health continue after childbirth and work to maintain her well-being. Reducing maternal death rates and improving overall maternal health requires effective risk management along the spectrum [1]. Maternal health is a global crisis, and every day, 830 women die from pregnancy or childbirth-related complications. Shockingly, 2.7 million newborns die within their first 4 weeks out of 130 million annual births. Accessible contraception and improved birth spacing can

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Engineering Reports* published by John Wiley & Sons Ltd.

reduce maternal mortality by 30% and child mortality by 20% (World Health Organization, 2021) [2]. Approximately 287,000 women died due to complications related to pregnancy and childbirth in 2020, with 95% of these deaths occurring in low and lower-middle-income countries. The majority of maternal deaths occur in Sub-Saharan Africa and Southern Asia, accounting for around 87% of the global total [3]. A growing focus has recently been on investigating automated and computerized methods for determining MHR. Thus, this research aims to build an automated system for MHR prediction by utilizing ML and XAI. ML is well known for providing accurate and easily understandable diagnostic information, making it a favored tool in medical research [4].

Several methods using ML techniques have been developed to address MHR in recent years. The study by Mutlu et al. [5] utilized a decision tree (DT) and achieved an accuracy range of 89.16%, focusing on risk prediction. Raza et al. [6] combined support vector machine (SVM) and DT-BiLTCN, achieved an impressive 98% accuracy, and suggested exploring user-friendly applications in future work. Mondal et al. [7] utilized random forest (RF) and proposed advancements in encryption for data privacy and enhanced security measures, achieving a 93.14% accuracy with the internet of medical things (IoMT) framework. Togunwa et al. [8] applied SVM, achieving a 95% accuracy in predicting MHR, and recommended incorporating additional variables for further optimization. Nirmala and Kambli [9] utilized RF in their ensemble model, achieved a 96% accuracy, and suggested using a hybrid model for increased accuracy. Ahmed and Kashem [10] used DT for IoT-based risk level prediction and achieved a 97% accuracy, respectively. In recent healthcare-focused ML research, XAI has been increasingly integrated to enhance transparency and trust in automated decision-making. For instance, Chadaga et al. [11] developed a stacked ensemble model for COVID-19 diagnosis using multiple deep learning algorithms. They employed five XAI techniques (SHAP, LIME, Eli5, QLatice, and Anchor) to interpret predictions, achieving an accuracy of 89%. Goenka et al. [12] proposed a deep learning approach using volumetric convolutional neural networks (ConvNet) for Alzheimer's disease detection, reaching an outstanding accuracy of 97%, with potential for visualization-based XAI integration in future work. Palkar et al. [13] utilized several ML and DL models, including RF, SVM, and XGBoost, for glioma prognosis; their XGBoost model achieved an accuracy of 88% while incorporating SHAP, LIME, Eli5, and QLatice for interpretability. Similarly, Vivek Khanna et al. [14] designed an ensemble stack model for detecting gestational diabetes mellitus, achieving a remarkable 96% accuracy, and employed multiple XAI methods to explain influential factors, such as visceral adipose tissue deposits. The analysis of these studies helps identify different research gaps and motivates us to conduct this research to rectify these issues.

This research utilizes an MHR dataset to identify factors influencing maternal health. It employs preprocessing techniques to refine the dataset and explore potential improvements in MHR identification using ML methods. Ten ensemble ML models are utilized to predict MHR in the dataset. These models are combined with feature optimization methods to improve results. The performance of each ML model on the dataset is evaluated using a range of performance metrics, both with and without feature optimization. After thoroughly considering all experiments,

the best outcome model is selected. The results of the chosen ensemble model are then analyzed using various XAI techniques, which significantly contribute to understanding the MHR factors. Figure 1 illustrates the complete pipeline of this study visually. It graphically presents the flow from data collection and preprocessing to feature optimization, model training and selection, and the integration of explainable AI. Additionally, it highlights the comparison between AI and clinical decisions, demonstrating how the final predictions can support personalized maternal health interventions. This figure provides a clear and concise visual representation of the research methodology.

The main contributions of this study are as follows:

- An automated maternal health risk prediction framework is proposed that combines ensemble ML with XAI, a combination that has not been jointly explored in many studies.
- A comprehensive evaluation of 10 ensemble models, along with three feature optimization techniques (PCA, LDA, and RFE), is conducted to determine the most effective model configuration systematically.
- The proposed Bagging + PCA model achieves the highest reported accuracy (99.51%) on this dataset, surpassing the performance of existing studies in the literature.
- Model interpretability is enhanced through the application of multiple XAI techniques (SHAP, LIME, FORCE, and ICE), addressing the lack of transparency in most previous works.
- The importance of features identified by the XAI models is validated through consultation with clinical experts, demonstrating strong alignment between AI-driven insights and medical expertise, thereby increasing the trustworthiness of the model.
- A practical deployment framework is proposed for clinical use, including real-world integration guidelines and solutions to implementation challenges in both urban and rural healthcare settings.

The paper is divided into several sections, each with a specific objective. Section 2 describes the research materials and methodology. Section 3 presents the outcome of this research. Section 4 focuses on the discussion associated with the outcome analysis of the proposed method. Finally, Section 5 concludes the study by summarizing the key findings and explaining their implications.

2 | Materials and Methodology

The primary objective of the research is to achieve the highest level of accuracy, along with associated explanations, in identifying maternal health patients using an ML technique. Figure 2 provides an overview of this research, and Sections 2.1–2.8 provide a detailed explanation of this figure.

2.1 | Data Collection

This research utilizes a dataset [15] comprising 1014 samples, each with seven features. Six of these features are predictive

Automated Maternal Health Risk Prediction using ML and XAI

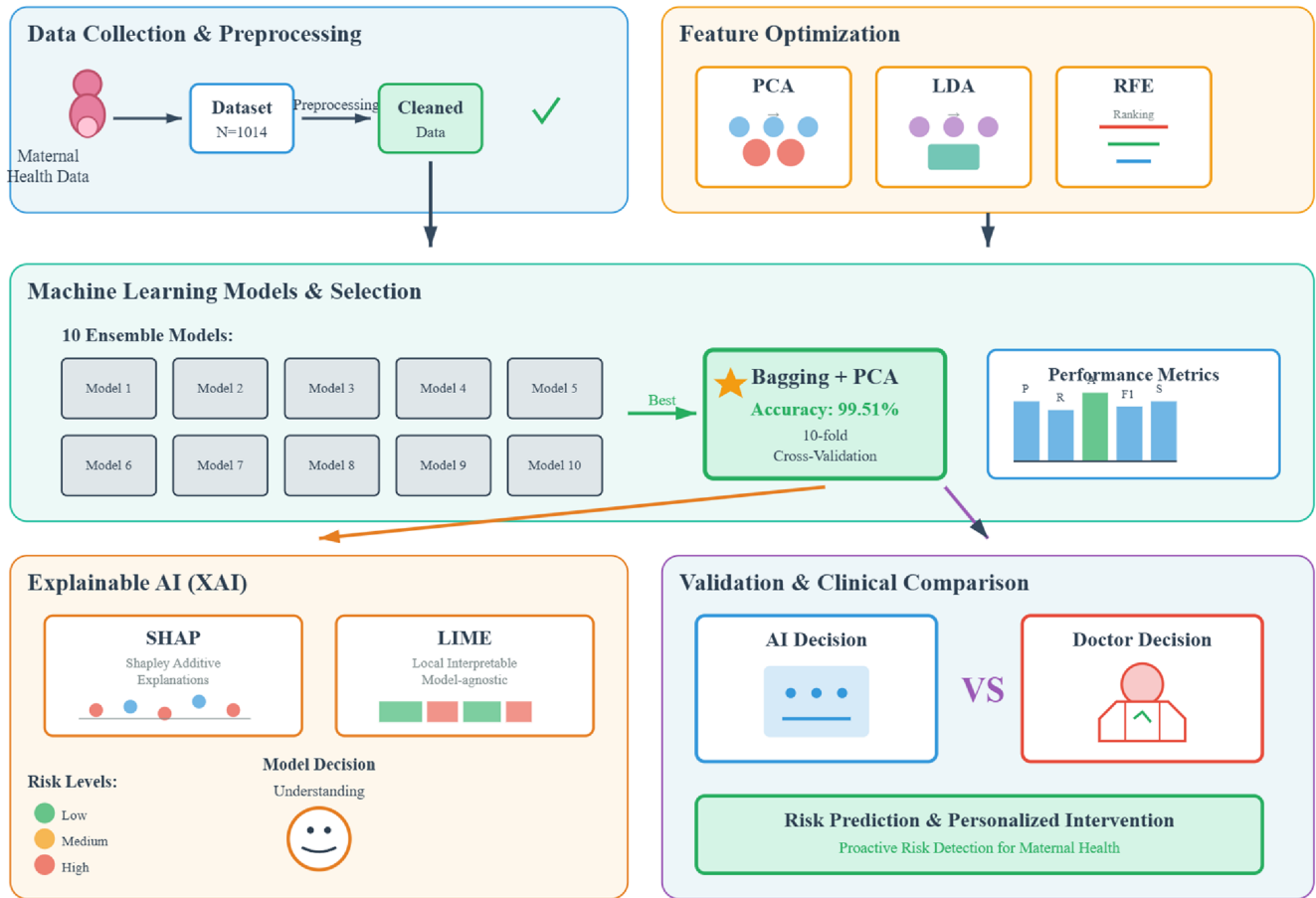


FIGURE 1 | Overall workflow of the proposed maternal health risk prediction framework.

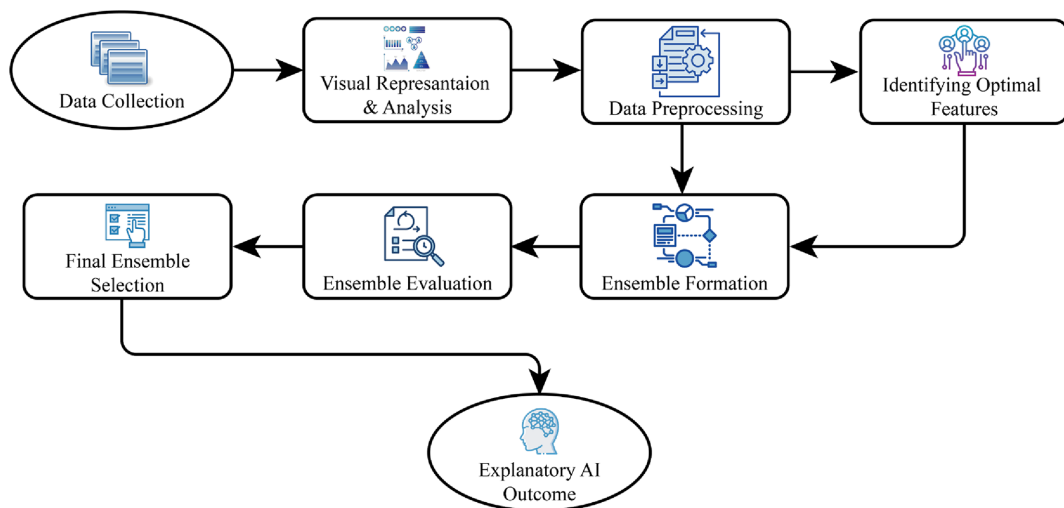


FIGURE 2 | Working principle of the proposed method.

variables, and one is the risk level (RL), which acts as the decision class. In this dataset, 406 samples are classified as low risk, 336 samples are associated with mid-risk, and 272 samples belong to high risk. The features chosen are Age, systolic blood pressure (SBP), diastolic blood pressure (DBP), blood sugar (BS), body

temperature (BT), and heart rate (HR). The dataset is categorized into low, mid, and high-risk groups based on specific maternal health parameters. Low risk is identified when the age falls between 20 and 29 years, systolic blood pressure (SBP) ranges from 120 to 139 mmHg, diastolic blood pressure (DBP) is between

TABLE 1 | Description of the features of the MHR dataset.

Features	Description	Value	Unit
Age	Age-related health risks.	Numeric	Years
SBP (SystolicBP)	Indicates the blood pressure in the arteries during a heartbeat, or the SBP.	Numeric	mmHg
DBP (DiastolicBP)	It indicates the diastolic blood pressure or arterial pressure during the resting phase of the heartbeat.	Numeric	mmHg
BS	Provides information on glucose regulation by reflecting the level of BS.	Numeric	mg/dL
BT (BodyTemp)	Represents body temperature, a gauge of a person's thermal status.	Numeric	°C
HR (HeartRate)	It displays the heart rate, which measures the health of the circulatory system.	Numeric	bpm
RL (RiskLevel)	Performs the role of the decision class, indicating, in light of the provided features.	Nominal	—

TABLE 2 | Characteristic list for the patient of the MHR dataset.

Parameter	Low risk	mid risk	High risk
Blood pressure	Systolic 120–139 mmHg, Diastolic 80–89 mmHg	Systolic 140–159 mmHg, Diastolic 90–99 mmHg	Systolic ≥ 160 mmHg, Diastolic ≥ 100 mmHg
Heart rate	75–80 bpm	90–140 bpm	> 70 and < 140 bpm
Body temperature	98.6°F (37°C)	$< 98.6^\circ\text{F}$ (37°C) and $> 102^\circ\text{F}$ (38.9°C)	$\geq 102^\circ\text{F}$ (38.9°C)
Fetal movement	10 movements within 12 h, 6 kicks/2 h	10 movements within 12 h, 6 kicks/2 h	> 10 movements within 12 h
Age	20–29	30–35	35–45
BMI	18.5–24.9 kg/m ²	Underweight: < 18.5 kg/m ² , Overweight: 25–29.9 kg/m ²	Obese: ≥ 30 kg/m ²
Blood glucose (2-h glucose)	< 7.8 mmol/L (< 140 mg/dL)	7.8–11.1 mmol/L (140–200 mg/dL)	≥ 11.1 mmol/L (≥ 200 mg/dL)

80 and 89 mmHg, blood sugar (BS) levels are under 7.8 mmol/L (140 mg/dL), body temperature (BT) is around 98.6°F (37°C), and the heart rate (HR) is between 75 and 80 bpm. Mid risk applies when the age is between 30 and 35, SBP ranges from 140 to 159 mmHg, DBP is between 90 and 99 mmHg, BS falls between 7.8 and 11.1 mmol/L (140–200 mg/dL), BT is below 98.6°F (37°C) or above 102°F (38.9°C), and HR ranges from 90 to 140 bpm. High risk is determined when the age is between 35 and 45, SBP is 160 mmHg or higher, DBP is 100 mmHg or higher, BS is 11.1 mmol/L (200 mg/dL) or more, BT exceeds 102°F (38.9°C) or shows hypothermia, and HR is over 140 bpm or below 70 bpm. Table 1 provides a comprehensive overview of the dataset features, ensuring a solid foundation for our analytical approach to this study.

The study involves pregnant women, particularly from rural areas in developing countries, who are monitored using IoT-enabled wearable devices to track health parameters. It incorporates data from external sources, including the Pima Indian Diabetes dataset [16], which serves as a reference for analyzing and determining risk factors (low, medium, high) in the MHR dataset. The classification of risk levels in the MHR dataset is informed by

insights gained from analyzing the Pima dataset, ensuring a more robust and evidence-based determination of risk levels. However, this study focuses on common pregnancy risk factors such as age, BMI, and blood glucose levels, while applying exclusion criteria to filter out non-pregnancy-related data and irrelevant health conditions. Table 2 summarizes the characteristics of patients contributing to the MHR dataset.

2.2 | Visual Representation and Analysis

This research analyzes several visual representations to observe the general patterns, outlier conditions, and other aspects of the MHR dataset. The visual representations include histograms with kernel density plots [17], violin plots [18], and heat maps [19]. The histogram is essential as it helps us understand the distributions and densities of the features in the dataset, allowing us to identify which features have the highest and lowest values in their distributions. The kernel density plot helps identify outliers, while the violin plot offers a more comprehensive view of the distribution, displaying both the probability density and the presence of outliers. To handle these outliers, we apply

standardization [20]. The heat map helps us discover the correlation among all dataset features. Understanding the correlation between features enables us to remove certain features during the RFE implementation method, retaining only those with positive and negative correlations, which helps the model better comprehend the dataset. Lastly, the scatter plot is an essential visualization technique for illustrating the relationship between the decision class feature and the other features in the dataset. These visualizations play a crucial role in identifying clusters and patterns within the dataset's components. By examining these relationships, we can gain insights into how different features interact and contribute to the overall prediction of MHR.

2.3 | Data Preprocessing

We utilize label encoding, an ML approach for converting categorical data into a numerical format, as a part of our preprocessing strategy. This method assigns a unique integer to each class, establishing an ordinal relationship among them. Label encoding is especially helpful when utilizing ensemble models that require numerical input, such as XGBOOST or RF. Our study employs an encoding method where low risk is represented by the numerical value 0, mid risk by 1, and high risk by 2. This encoding schema is structured such that the assigned values correspond to the perceived magnitude of risk, with 0 indicating the lowest risk, 1 representing a moderate level of risk, and 2 denoting the highest level of risk in our scenario. Additionally, the research uses standard scalarization during preprocessing to ensure that all features derived from the encoded risk levels have a mean of 0 and a standard deviation of 1, contributing to more consistent model performance.

2.4 | Identifying Optimal Features

Optimal feature selection in ML involves choosing the most informative variables from a dataset to enhance model performance and reduce time complexity while executing ML models. This research analyzes three feature optimization techniques, namely LDA, PCA, and RFE, with the ML models. LDA is a method in statistics and ML that aims to identify linear combinations of features, called discriminants, to maximize the separation between multiple classes. It is commonly used for dimensionality reduction and classification tasks, finding the optimal projection of data points onto a lower-dimensional space. PCA is a widely used technique for feature extraction and dimensionality reduction. By converting the data into a new coordinate system, PCA highlights the most vital information and lowers the dataset's dimensionality. RFE is a feature selection method in ML that systematically removes the least significant features from a dataset [21]. These methods help reduce the dimensionality of the data by selecting the most relevant features, thereby enhancing the efficiency of ML models. Specifically, we use three principal or discriminant components for PCA and LDA to ensure that most of the data's variance is captured while minimizing the loss of information. By applying these feature optimization techniques, we aim to improve the accuracy and execution speed of the ML models under consideration. This research uses feature optimization techniques on both training and testing datasets.

2.5 | Ensemble Formation

This research on MHR involves an exhaustive application of ensemble models, incorporating grid search and meticulous hyperparameter tuning, coupled with L1 and L2 regularization techniques [22]. We deploy a total of 10 ensemble models, including extreme gradient boosting (XGBOOST) [23], RF [24], bootstrap aggregating (Bagging) [25], boosting [26], stacking [27], voting [28], gradient boosting machine (GBM) [29], light gradient boosting machine (LGBM) [30], gradient boosting decision tree (GBDT) [31], and random patches (RP) [32] with different feature optimizations as PCA, LDA and RFE. The evaluation revealed that the bagging ensemble method, synergized with PCA for feature optimization, is the most effective for MHR assessment.

2.5.1 | XGBOOST

XGBOOST is an efficient and scalable implementation of gradient boosting. XGBOOST uses a boosting technique to combine weak learners (decision trees) into a strong learner. It iteratively corrects the errors made by the previous model and has gained popularity due to its performance in various ML tasks. XGBOOST includes the learning rate, which controls the contribution of each tree to the final model. Lower values prevent overfitting but require a larger number of trees. The number of estimators represents the total number of trees in the model, while the maximum depth specifies the maximum depth of each decision tree. Regularization is controlled using Lambda (L2 regularization) and Alpha (L1 regularization), which are used to avoid overfitting and control sparsity [33].

2.5.2 | RF

RF is an ensemble learning method that combines multiple decision trees trained on random subsets of the data. The final prediction is made by aggregating the predictions of individual trees through voting (for classification) or averaging (for regression). RF includes the number of trees, which indicates the number of trees in the forest. The maximum depth of each tree is set to prevent overfitting, and the minimum samples per leaf specify the minimum number of samples required to be at a leaf node. Additionally, the maximum features hyperparameter determines the number of features to consider when splitting a node [34].

2.5.3 | Bagging

Bagging is an ensemble method that builds multiple models (typically decision trees) on different bootstrap samples of the dataset. Predictions are then averaged or voted upon. In Bagging, the number of estimators indicates the number of base learners (models) included in the ensemble. The maximum features parameter specifies the number of features considered when training each model. The random state hyperparameter controls the randomness of data sampling in each model to ensure reproducibility [35].

2.5.4 | Boosting

Boosting is an ensemble method that focuses on iteratively correcting the errors made by previous models. Each model in the sequence is trained to improve the performance of the model that came before it. The main parameters for Boosting include the learning rate, which determines how much each new model contributes to the final prediction. The number of estimators refers to the total number of boosting rounds (iterations) in the model. The base learner used in boosting is typically a decision tree that serves as the weak learner [36].

2.5.5 | Stacking

Stacking is a meta-learning ensemble technique where multiple base models are trained, and their outputs are combined using another model (the meta-model). The meta-model learns to predict the final output based on the predictions of the base models. In Stacking, the base models are the individual models that make up the ensemble. The meta-model is the model used to combine the predictions of base models. The number of estimators indicates the number of base models that form the ensemble [37].

2.5.6 | Voting

Voting is an ensemble method that combines predictions from multiple models using either majority voting (for classification) or averaging (for regression). It is a simple yet effective method of improving model performance by leveraging the diversity of models. The voting type parameter can be “hard” for classification (majority voting) or “soft” for regression (averaging the predictions). The individual models refer to the set of models contributing to the ensemble’s final decision [38].

2.5.7 | GBM

GBM is an ensemble method that builds models sequentially, each one correcting the errors of the previous one. It is a powerful technique, particularly for structured data. For GBM, the learning rate determines the extent to which each new model contributes

to the final prediction. The number of estimators specifies the number of boosting rounds or trees in the model. The maximum depth of the trees is another critical parameter. The subsample hyperparameter determines the sample fraction that is used for each base learner [39].

2.5.8 | LGBM

LGBM is a faster and more memory-efficient implementation of gradient boosting. It is optimized for large datasets, providing superior training time and accuracy. LGBM’s key parameters include the learning rate, which controls the contribution of each model to the final prediction, and the number of estimators, which specifies the number of boosting rounds. The number of leaves is the number in each tree, while the maximum depth sets the maximum depth of each tree in the model [40].

2.5.9 | GBDT

GBDT is another variant of gradient boosting that utilizes decision trees as weak learners. It is known for its high predictive power and flexibility. The main parameters for GBDT include the learning rate, which controls the contribution of each tree, and the number of estimators, which specifies the total number of boosting rounds. The tree depth controls the maximum depth of the individual decision trees [41].

2.5.10 | RP

RP is an ensemble technique that works similarly to Bagging but uses random subsets of data and features for each model in the ensemble. This increases diversity among the individual models and can improve performance, including the number of estimators, which specifies the number of models to be included in the ensemble. The sample size determines the data points used for each model, and the feature proportion sets the fraction of features used for each model [42].

Table 3 provides a concise overview of key hyperparameters for various ensemble models. The learning rate (LR) controls the

TABLE 3 | Hyperparameters of models.

Model	LR	NE	MD	λ	α	NT	MSPL	MF	SS	NL	SSZ	FPR
XGBOOST	0.05	100	6	1.0	0.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A
RF	N/A	200	12	N/A	N/A	200	10	0.8	N/A	N/A	N/A	N/A
BAGGING	N/A	150	10	N/A	N/A	150	5	0.9	N/A	N/A	N/A	N/A
BOOSTING	0.1	50	8	N/A	N/A	N/A	N/A	N/A	0.8	N/A	N/A	N/A
STACKING	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
VOTING	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
GBM	0.05	100	6	N/A	N/A	N/A	N/A	N/A	0.8	N/A	N/A	N/A
LGBM	0.1	100	8	N/A	N/A	N/A	N/A	N/A	0.9	50	N/A	N/A
GBDT	0.05	120	6	N/A	N/A	N/A	N/A	N/A	0.7	N/A	N/A	N/A
RP	N/A	100	8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.7	0.8

TABLE 4 | Description of performance measurement metrics.

Name	Equation	Meaning
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN} \times 100$	Total accuracy of the MHR prediction.
Precision	$\frac{TP}{TP+FP} \times 100$	Percentage of correctly identified cases of MHR.
Recall	$\frac{TP}{TP+FN} \times 100$	Accurate identification of risks to maternal health.
F1 Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100$	Durability of ensemble models in assessing the risk to maternal health.

step size during the optimization process. The number of estimators (NE) represents a model's total iterations or trees. Maximum depth (MD) limits the depth of each tree, preventing overfitting. Lambda (λ) and alpha (α) correspond to L2 and L1 regularization, respectively, helping reduce model complexity. The number of trees (NT) refers to the number of trees used in tree-based methods. Minimum samples per leaf (MSPL) indicates the minimum samples required to create a leaf node. Maximum features (MF) defines the subset of features considered during the split selection process. Subsample (SS) denotes the fraction of data used for training in boosting methods. Number of leaves (NL) specifies the leaf count in LGBM. Sample size (SSZ) and feature proportion (FPR) refer to the size of samples and the fraction of features used in random projection-based methods. These parameters are critical for tuning models to achieve optimal performance.

2.6 | Ensemble Evaluation

This research uses ML performance metrics such as confusion matrix (CM), accuracy, precision, recall, F1 score, and ROC curve to evaluate the model performance. A CM is a graphical representation that illustrates the performance of a classification ensemble model, with 'accurate' and 'false' indicating actual and predicted outcomes, respectively. CM consists of four entries: False positive (FP), false negative (FN), true positive (TP), and true negative (TN) values of different classes [43]. Together, these measures provide a comprehensive evaluation of the model's ability to distinguish between various degrees of MHR. We include a CM in the results section to assess the predictive performance of our various models effectively. Table 4 illustrates the measurement of different performance metrics. The ROC curve graphically represents the binary classification performance of our model in assessing MHR. It demonstrates the trade-off between the y-axis's actual positive rate (TPR) and the x-axis's false positive rate (FPR) across different classification thresholds.

All experiments are assessed through 10-fold cross-validation. Fold cross-validation splits the dataset into k ($k=10$ for this research) subsets, training the model on $k-1$ folds and validating it on the remaining fold, repeating this for all folds. This technique provides a robust estimate of the model's performance by averaging the results across all iterations.

In our approach, the dataset is initially split into 80% for training and 20% for testing. Within the training portion, we perform iterative folding, where each fold involves using a different subset of the 80% as the training set. In contrast, the remaining portion serves as the validation set. This process is repeated across all folds, ensuring that every part of the 80% is used for training and validation. Meanwhile, the 20% test set remains fixed throughout the process, providing a consistent basis for final evaluation. This method allows robust model training and validation while maintaining an unbiased performance assessment on the held-out test set [44].

2.7 | Final Ensemble Selection

In our research, we conduct 10 ensemble models, assessing their performance using various metrics. Subsequently, we apply several feature optimization techniques, which lead us to select bagging as the most effective ensemble model for our maternal health dataset with feature optimization using PCA. Hence, bagging with PCA is the final model for identifying MHR.

2.8 | Explanatory AI Outcome

This study employs multiple XAI to explain its outcomes. Various strategies are employed in the world of XAI to generate human-accessible explanations of AI predictions and methods for understanding the internal mechanisms and potential biases of AI models. In this research, XAI tools such as SHAP plots and LIME plots are employed to enhance our understanding of AI decision-making processes and ensure that the outcomes are valid and comprehensible. SHAP is a popular technique of XAI. It clarifies ML model outputs by allocating weights for each feature's impact on the prediction. Determining SHAP values provides a comprehensive and consistent understanding of how individual features contribute to the model's output. Another essential tool in XAI, LIME is a crucial technique that generates explanations based on the coefficients of the understandable model by observing the complex model's response to perturbed instances [45]. Furthermore, to gain a deeper insight into how individual instances behave with respect to each feature, we also employed ICE plots for the best-performing model. ICE plots visualize how the prediction for a single sample changes when a specific feature varies, allowing us to capture instance-level variations and uncover heterogeneous effects that may be hidden in global explanations [46].

3 | Result and Discussion

Figure 3 offers a comprehensive insight into the characteristics of MHR data through combined histograms and a density plot, visually representing the distribution of the dataset. The histograms organize values into intervals, with the heights of the bars indicating the frequencies of the data. At the same time, the density plot illustrates the conditioned probability density function of the continuous dataset, revealing distinct distribution patterns across different parameter ranges. Together, these visualizations are potent tools in statistics and ML, adept at highlighting outliers and distribution features, thus assisting in comprehensive

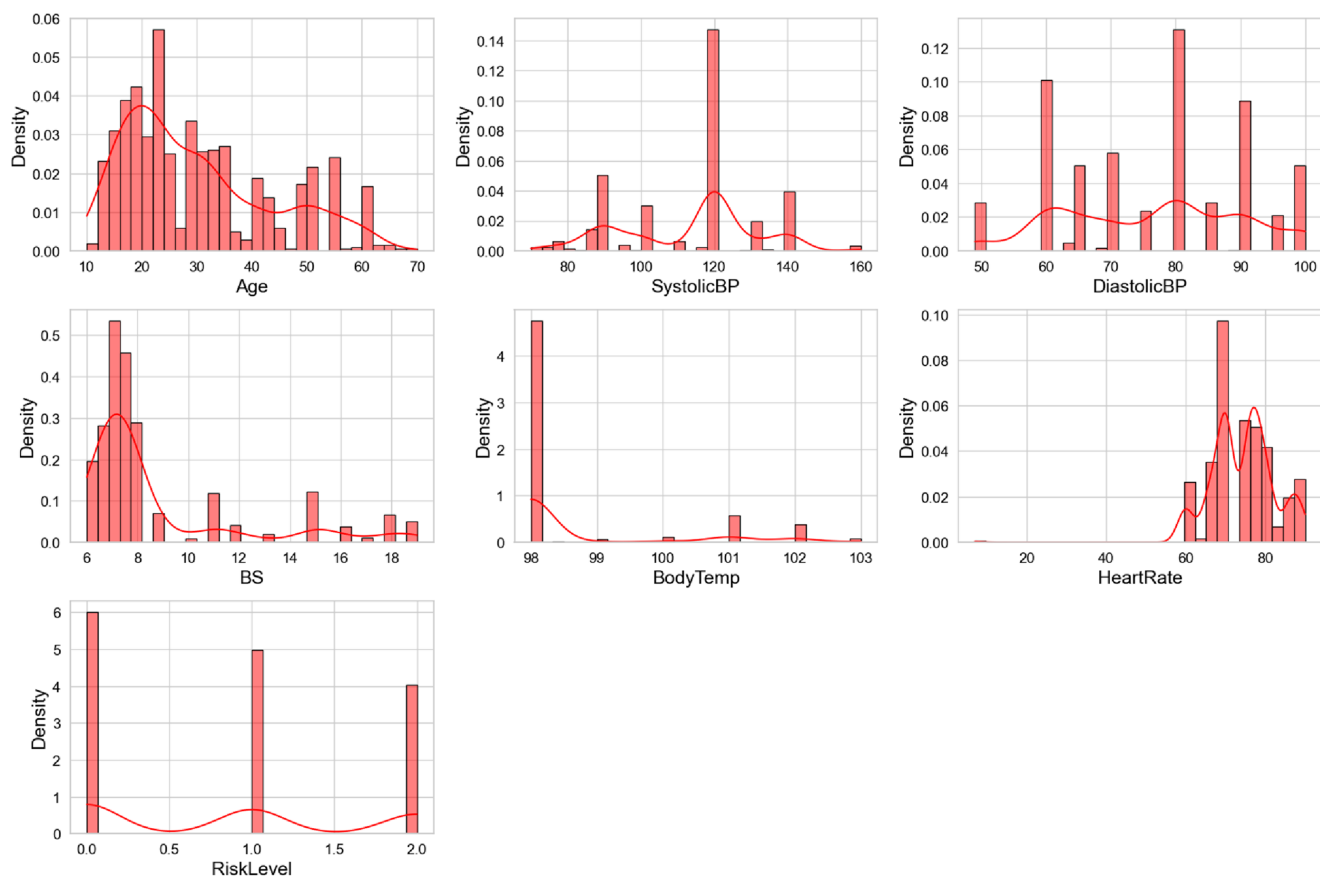


FIGURE 3 | Histogram with kernel density of different features.

data perception. Within the MHR dataset, the histograms reveal concentrated values within specific ranges across most columns, likely indicating distinct features that aid in assessing MHR. This concentration provides a clear picture of the distribution, guiding data analysis and informing modeling strategies. Additionally, the density plot demonstrates that the distributions of BS, BT, and HR dominate at lower levels within their respective ranges, indicating a concentration of data points in these regions. In contrast, Age, SBP, and DBP exhibit high-density values, reflecting a broader and more evenly distributed spread across their ranges. This combined visualization enhances the understanding of the dataset's characteristics and potential trends, with specific parameters showing concentrated distributions while others display more uniform spreads.

Figure 4 shows the violin plots, a visualization tool that combines elements of a box plot and a kernel density plot to demonstrate the variance of a constant variable across various classifications. Its dimensions, which resemble those of a violin, reflect the density of the data. These plots illustrate the distribution of a persistent variable across segments within a dataset. Each class is depicted as a violin shape, containing a kernel density plot and nested box plot elements, facilitating efficient comparison among the different segments. This visualization method provides a comprehensive view of the data distribution, enabling insights into variance and patterns across various classifications.

Figure 5 displays heatmaps, a data visualization technique that utilizes colors to represent individual data points within a

grid structure, thereby converting structural information into a graphical format. These heatmaps, designed explicitly for the MHR dataset, elucidate connections and patterns among diverse datasets, uncovering differences and similarities. By employing color gradients, the heatmaps highlight variations in data values across different parameters or categories, facilitating the identification of trends, correlations, and potential insights within the dataset. This visualization method is a powerful tool for data exploration and analysis, enabling the interpretation and understanding of complex data.

All experiments in this research were conducted using a 10-fold cross-validation approach. We partition the dataset in an 8:2 ratio for training and testing the models in each fold. The models' performance is assessed based on the evaluation parameters described in Section 2.6. Table 5 presents the overall performance of various approaches, both with and without feature optimization, in this research. It shows that the PCA with the bagging model reached the highest accuracy of 99.51% among all the ensemble models. Additionally, with RFE, the VOTING ensemble model exhibits the lowest accuracy at 76.42%. Therefore, PCA with bagging is the final model to identify MHR in this research.

Table 6 presents the evaluation results of the Bagging + PCA model, which was selected as the best-performing model after assessing 10 algorithms. It outlines the performance metrics, including Accuracy, Precision, Recall, and F1, across each of the 10 folds in cross-validation. Alongside the individual fold values, the table includes the mean for each metric, reflecting the

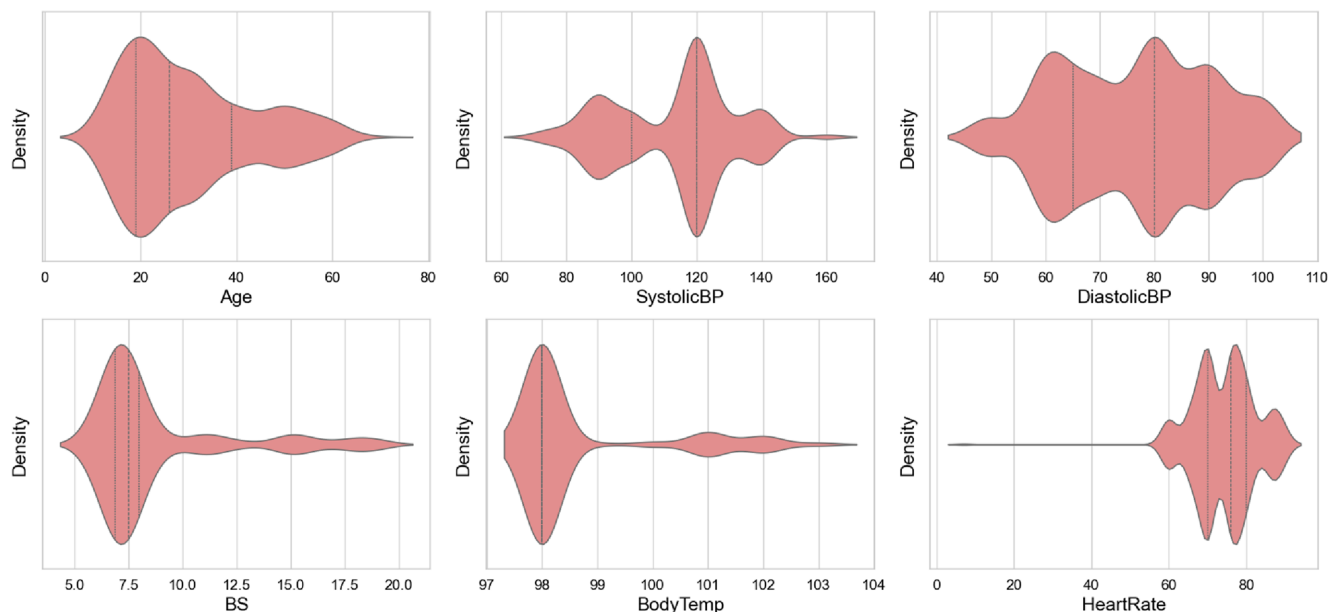


FIGURE 4 | Violin plot of different features.

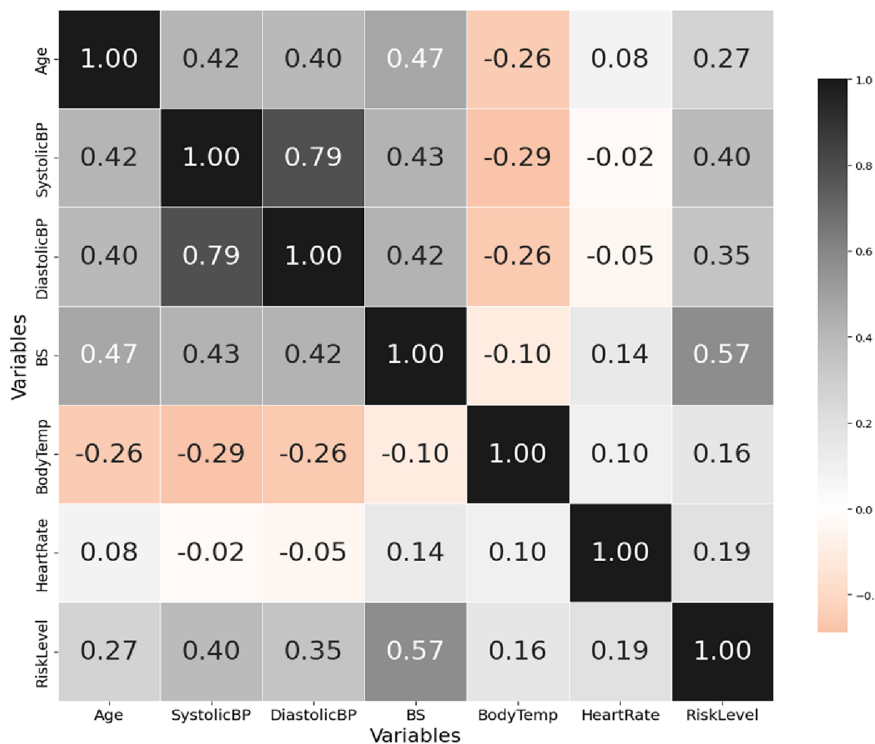


FIGURE 5 | Correlation heatmap of different features.

average model performance, the standard deviation (SD), which indicates the variability and consistency of the metric across the folds [47], and the confidence interval (CI), which provides a range within which the actual metric value is likely to fall with 95% confidence [48]. The results demonstrate that the Bagging + PCA model exhibits strong, stable performance with minimal fluctuation, making it a reliable choice for the evaluation task.

After evaluating all the models, we found that Bagging combined with PCA achieved the highest accuracy. We generated all XAI

methods for this combination to enhance understanding and provide clearer insights. Additionally, we applied XAI using only Bagging, without feature engineering, to better understand the impact of feature optimization on the model's predictions. This approach offers a more comprehensive interpretation of the model's behavior. Figure 6 presents the SHAP values for the proposed method, highlighting the influence of various features on model predictions in the context of MHR. We applied SHAP plots to our Bagging model in two distinct scenarios: (a) without feature optimization and (b) with feature optimization

TABLE 5 | Performance of different models with and without feature optimization.

Ensemble model	Accuracy	Precision	Recall	F1
XGBOOST	98.57	98.79	98.57	98.54
RF	97.14	97.67	97.14	97.13
BAGGING	98.57	98.83	98.57	98.56
BOOSTING	90.00	91.52	90.00	89.70
STACKING	97.85	98.24	97.85	97.84
VOTING	95.71	96.44	95.71	95.65
GBM	93.57	94.63	93.57	93.47
LGBM	98.57	98.85	98.57	98.58
GBDT	99.28	99.42	99.28	99.29
RP	93.57	94.68	93.57	93.48
XGBOOST+PCA	96.42	96.98	96.42	96.34
RF+PCA	96.42	97.00	96.42	96.34
BAGGING+PCA	99.51	98.99	99.51	99.67
BOOSTING+PCA	87.85	89.76	87.85	87.69
STACKING+PCA	95.71	96.41	95.71	95.63
VOTING+PCA	92.14	93.13	92.14	91.96
GBM+PCA	89.28	91.02	89.28	89.17
LGBM+PCA	94.94	95.98	94.94	94.95
GBDT+PCA	96.42	97.00	96.42	96.33
RP+PCA	92.08	92.92	92.08	91.98
XGBOOST+LDA	96.42	97.04	96.42	96.39
RF+LDA	95.00	95.88	95.00	94.94
BAGGING+LDA	95.00	95.88	95.00	94.94
BOOSTING+LDA	88.57	90.86	88.57	88.18
STACKING+LDA	95.71	96.39	95.71	95.61
VOTING+LDA	94.28	95.18	94.28	94.16
GBM+LDA	92.14	93.32	92.14	92.11
LGBM+LDA	95.00	95.85	95.00	94.95
GBDT+LDA	96.42	97.04	96.42	96.40
RP+LDA	94.28	95.26	94.28	94.14
XGBOOST+RFE	86.42	87.26	86.42	86.33
RF+RFE	85.00	86.36	85.00	85.08
BAGGING+RFE	85.00	86.38	85.00	84.98
BOOSTING+RFE	83.57	85.23	83.57	83.33
STACKING+RFE	85.00	86.34	85.00	84.96
VOTING+RFE	76.42	78.78	76.42	75.07
GBM+RFE	86.42	87.08	86.42	86.31
LGBM+RFE	85.71	86.90	85.71	85.76
GBDT+RFE	86.42	87.70	86.42	86.41
RP+RFE	87.14	88.62	87.14	87.19

Note: The bold value represents highest performance in this study, and it has been highlighted to help readers easily identify and compare the results.

using PCA. Both plots were generated to visualize the impact of the features in each case. This allows us to observe how feature selection through PCA enhances the model’s interpretability and performance. By comparing the results, we gain deeper insights into the importance of features with and

without PCA, showcasing how each approach influences the model’s decisions.

Figure 7 displays the LIME plot of the proposed method, illustrating the contribution of features to the model’s predictions.

TABLE 6 | Bagging + PCA model performance with mean, standard deviation, and confidence intervals.

Fold/metric	Accuracy	Precision	Recall	F1
1	99.51	98.97	99.51	99.64
2	99.50	98.97	99.50	99.78
3	99.53	99.00	99.53	99.75
4	99.53	98.89	99.53	99.62
5	99.50	98.90	99.50	99.71
6	99.50	98.96	99.50	99.63
7	99.58	98.94	99.58	99.68
8	99.54	99.35	99.54	99.59
9	99.45	98.97	99.45	99.60
10	99.49	98.99	99.49	99.69
Mean	99.51	98.99	99.51	99.67
SD	0.03	0.13	0.03	0.06
CI	(99.48, 99.53)	(98.90, 99.08)	(99.48, 99.53)	(99.62, 99.71)

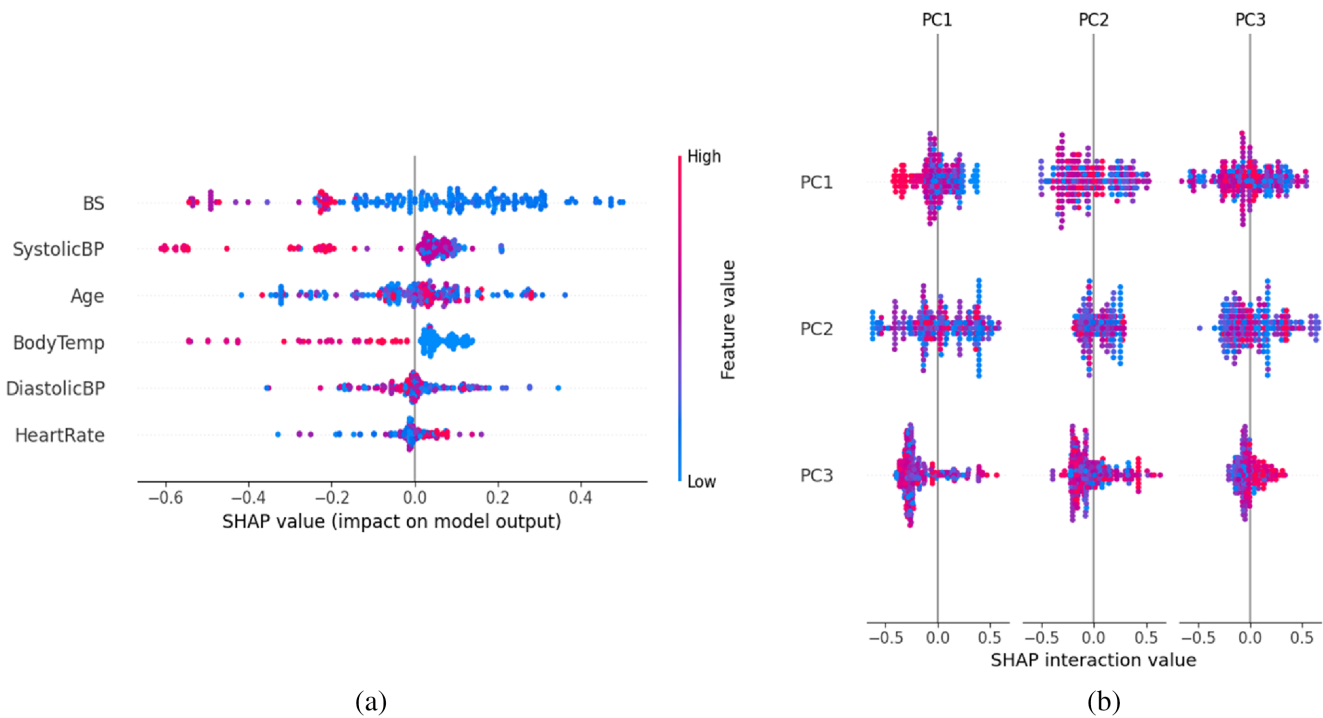


FIGURE 6 | SHAP plot of the proposed method: (a) without feature optimization and (b) with PCA.

The figure compares two scenarios: (a) without feature optimization and (b) with feature optimization using PCA. In the plot, each horizontal bar represents the influence of a feature on the model's prediction, with the length of the bar indicating its importance. This allows for a visual comparison of how the features impact the model's predictions in both cases, highlighting the effect of applying PCA versus not using it. The comparison offers additional insight into how feature selection impacts the model's decision-making process.

Figure 8 presents the mean SHAP plot for MHR, providing a clear and insightful overview of attribute values, organized by their mean influence as determined by SHAP values. The

figure compares two scenarios: (a) without feature optimization and (b) with feature optimization using PCA. In both plots, the features are systematically arranged based on their mean influence, effectively highlighting the key features that significantly impact the model's predictions related to maternal health risks. By presenting the attributes in a structured manner according to their influence, these plots enhance our understanding of each feature's importance and role in shaping the model's decisions. This graphical representation facilitates the identification of the most influential factors, offering valuable insights into the elements that significantly impact the model's predictions regarding MHR, both with and without feature optimization.

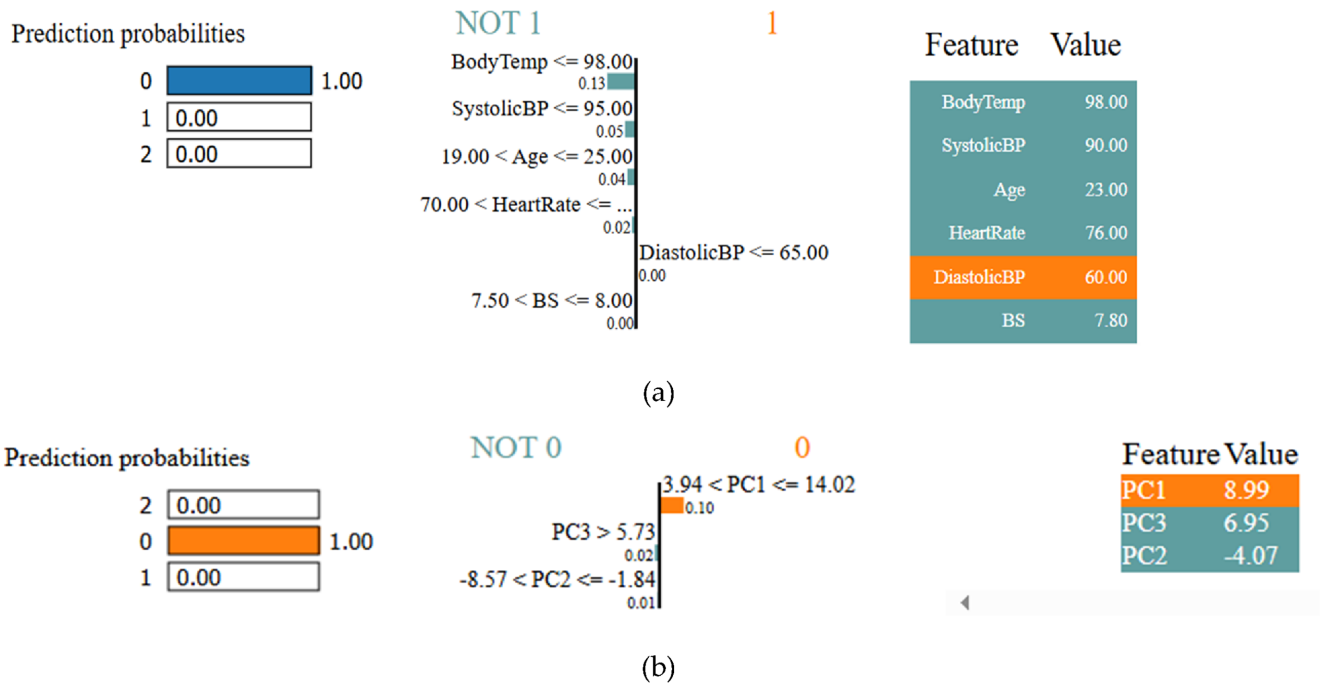


FIGURE 7 | The LIME plot of the proposed method: (a) without feature optimization and (b) with PCA.

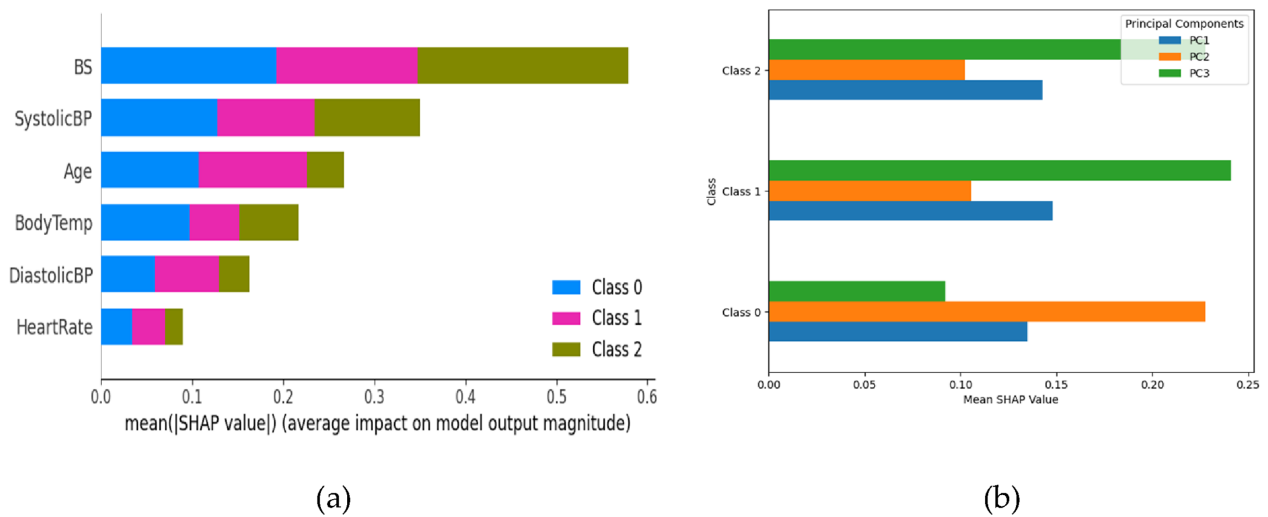


FIGURE 8 | Mean SHAP plot of the proposed method: (a) without feature findings and (b) with PCA.

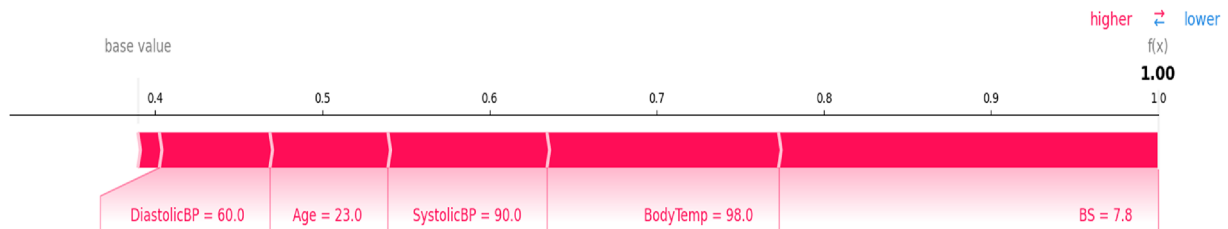


FIGURE 9 | FORCE plot of the proposed method.

Figure 9 represents a force plot, intertwined with SHAP values, and visually depicts the impact of individual features on a model's prediction in the context of MHR. Starting with the base prediction, colored bars represent the contribution of each

feature, with positive bars indicating an increase and negative bars indicating a decrease in the prediction. This makes it easier to interpret critical features and their specific influence on the model's output in the context of MHR.

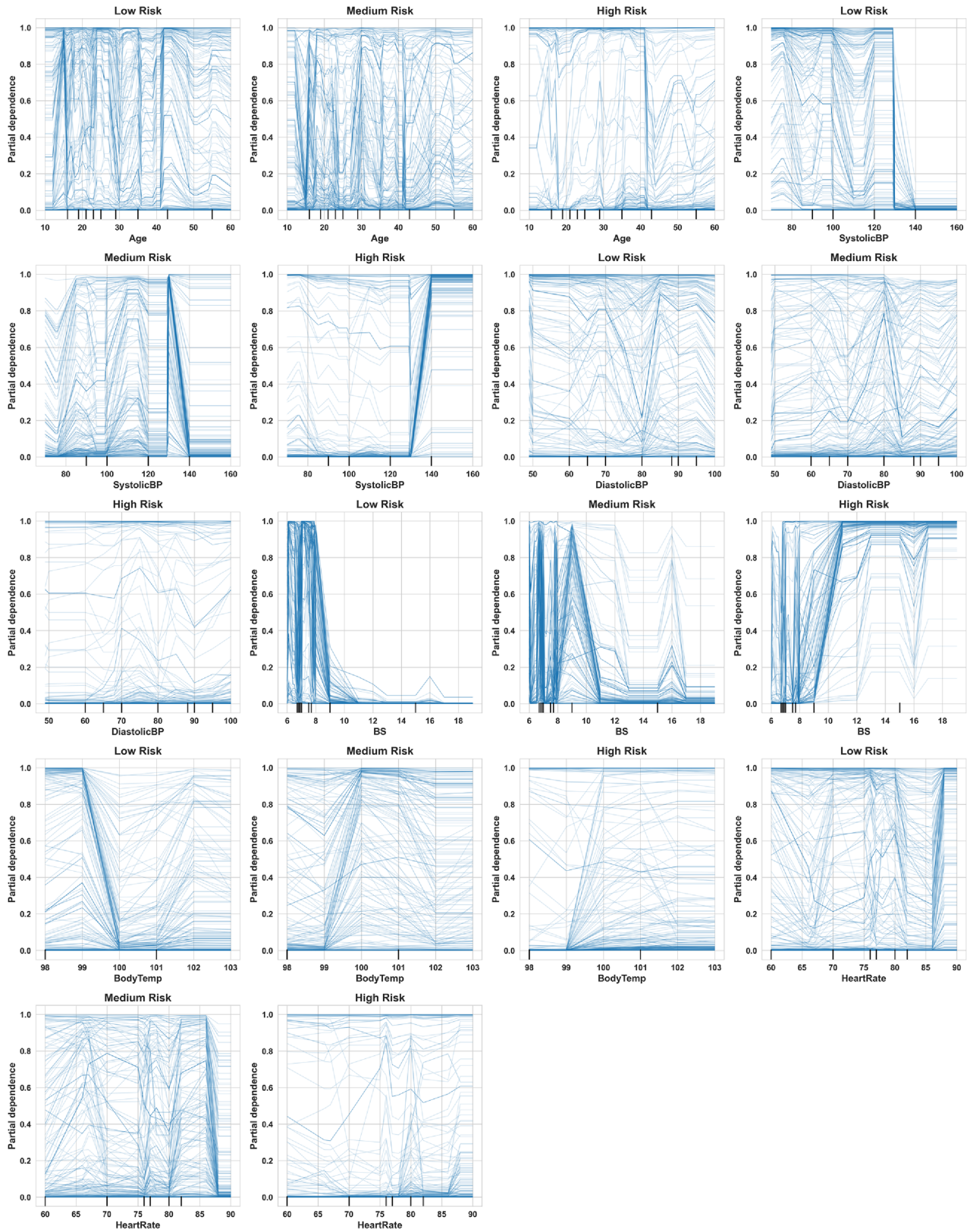


FIGURE 10 | Individual conditional expectation analysis of the proposed technique for MHR stratification without feature findings.

ICE plots depicting the localized effect of six clinical predictors: Age, SystolicBP, DiastolicBP, body temperature (BodyTemp), blood sugar (BS), and heart rate (HeartRate) on the predicted MHR levels (low, medium, and high) as learned by the Bagging

ensemble model. Each curve represents the change in predicted probability for a single instance as one feature varies while all others are held constant. The ICE plots in Figure 10 highlight distinct nonlinear and heterogeneous feature response patterns

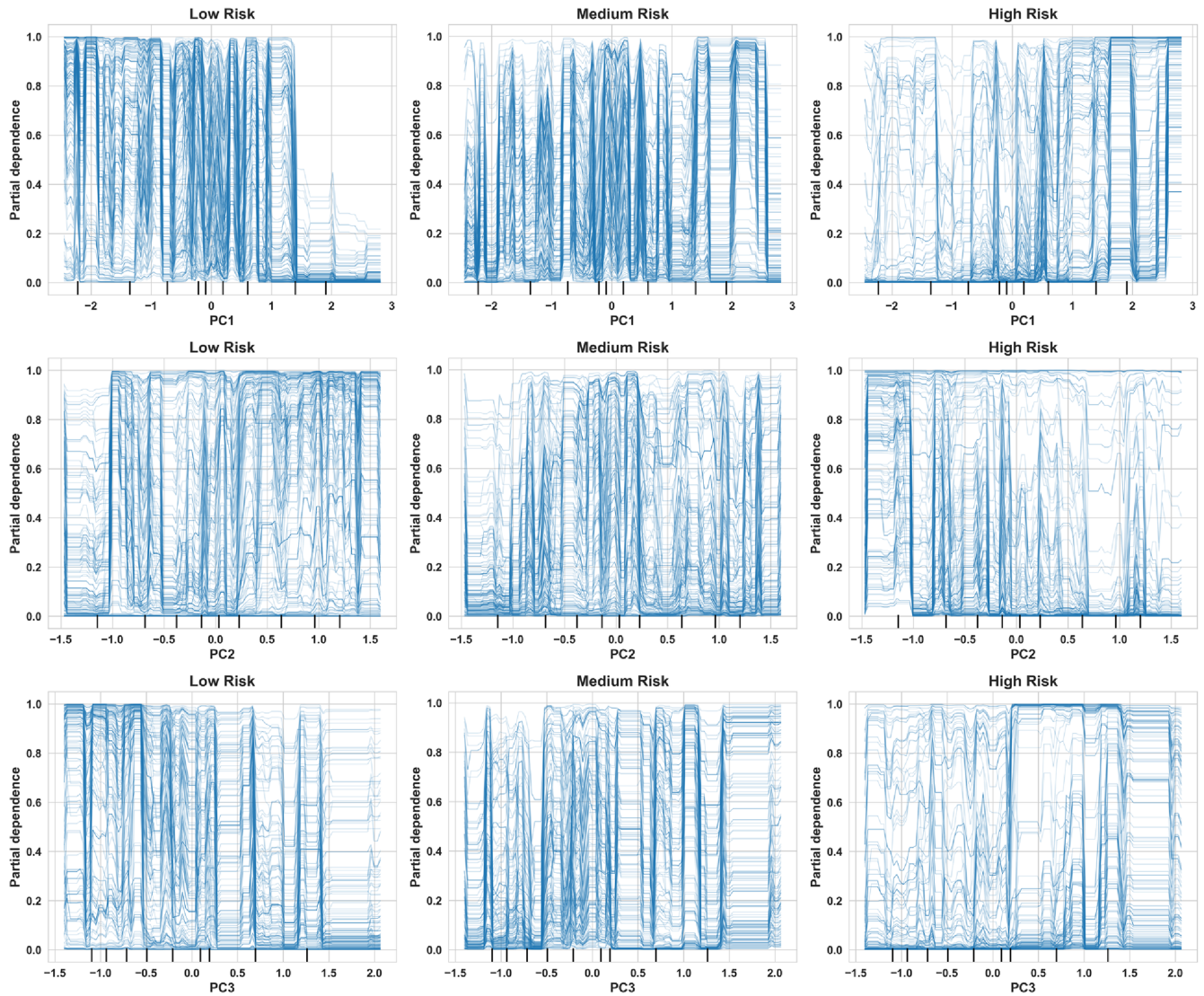


FIGURE 11 | ICE analysis of principal components for MHR Prediction using the proposed technique.

captured by the Bagging model without feature findings. *Age* and *SystolicBP* exhibit strong monotonic trends where increasing values are consistently associated with elevated *High Risk* probabilities and declining *Low Risk* likelihoods. *Blood Sugar* and *Body Temperature* demonstrate moderate positive associations with *Medium* and *High Risk* predictions, reflecting their physiological relevance in maternal complications. In contrast, *Heart Rate* and *DiastolicBP* show more dispersed and individualized trajectories, indicating subject-specific variability. Collectively, these plots confirm that the Bagging ensemble effectively models complex nonlinear interactions across maternal health indicators, supporting its interpretability and reliability in risk stratification.

Figure 11 illustrates the ICE plots of the proposed technique, based on the three principal components, which provide a transformed view of the model's decision landscape by combining correlated clinical features into orthogonal dimensions. PC1 demonstrates a clear separation between Low- and High-Risk regions, indicating that this component captures the most dominant variance associated with elevated maternal health risk.

PC2 and PC3 further refine this distinction by revealing localized variations in predicted risk probabilities, suggesting that complex nonlinear feature interactions are embedded within these components. Across all three principal components, the Bagging model exhibits distinct and consistent probability shifts, confirming its ability to generalize effectively over the latent feature space derived through PCA. This analysis highlights that the ensemble not only performs reliably on raw clinical attributes but also maintains interpretability and stability when applied to dimensionality-reduced representations, reinforcing its robustness in maternal health risk stratification.

Figure 12 presents the ROC curves for MHR. The bagging model, selected as the final model with PCA, achieves an accuracy of 99.51%. Figure 13 displays the CM, which represents the classification of three classes: Low Risk, Mid Risk, and High Risk. The model accurately classifies 80.8% of instances as Low Risk, 67.0% as Mid Risk, and 54.0% as High Risk. Misclassifications are minimal, with 0.2% of Low-Risk instances misclassified as Mid Risk, 0.4% as High Risk, and a similarly low proportion of misclassifications across other classes. The model demonstrates high

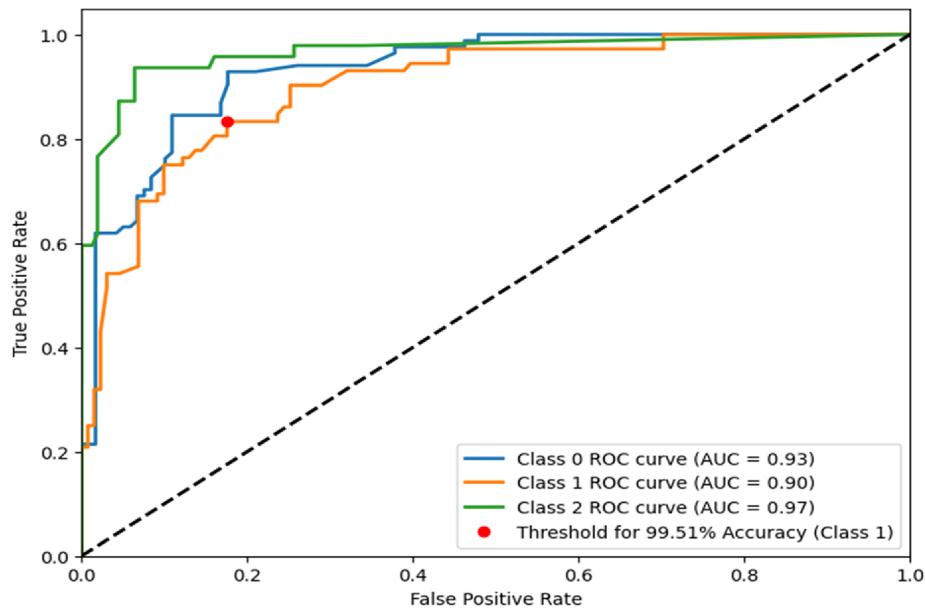


FIGURE 12 | ROC curve for the proposed technique.

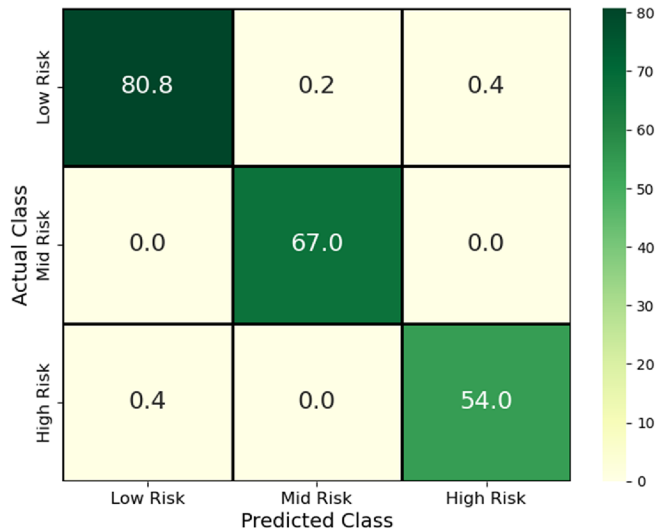


FIGURE 13 | CM of the proposed technique.

accuracy and reliability, with minimal errors across all categories. The ROC curve and CM are generated based on the average values from 10-fold cross-validation, enhancing the reliability and interpretability of the model's performance. This evaluation approach is crucial for establishing trust in AI systems, particularly in critical fields such as healthcare, where accurate predictions are vital for informed decision-making [49].

4 | Discussion

Table 7 compares the feature importance rankings from XAI models SHAP, LIME, and FORCE with clinical expert suggestions. The table shows a strong alignment between the models and experts, particularly in their top priority: SHAP and FORCE, along with one of the clinical experts, identify Blood Sugar as the most critical factor. Additionally, the XAI models and the

experts emphasize the significance of blood pressure (Systolic and diastolic), ranking Systolic BP highly, and the experts note blood pressure as a critical risk factor. This convergence between AI-driven insights and expert judgment, particularly in key features such as blood sugar and blood pressure, is a promising outcome that suggests XAI models are making decisions closely aligned with clinical expertise, thereby supporting their reliability in healthcare analysis.

Table 8 compares our proposed method with existing research. It demonstrates that our approach makes a significant contribution across all methods, particularly in terms of accuracy and XAI.

Furthermore, several recent studies have explored MHR prediction using different ML techniques. Togunwa et al. [8] employed ANN and RF for maternal health classification, achieving a 95% accuracy rate. Abdulla-Hil-Kafi et al. [50] achieved 98% accuracy with RF for MHR prediction, and Thakkar et al. [51] reported 94.26% accuracy using the same method for forecasting MHR. Bajaj et al. [52] explored various ML algorithms for risk level prediction, though without reporting accuracy, and Maheswari et al. [53] applied XGBoost with hyperparameter tuning, incorporating SHAP, LIME, and SMOTE, achieving 88.89% accuracy. Only Maheswari et al. employed XAI among these studies, while the rest did not implement these critical aspects. Additionally, recent healthcare-related AI research has increasingly focused on integrating XAI to enhance interpretability and clinical trust. Chadaga et al. [11] developed a stacking ensemble for COVID-19 diagnosis using SHAP, LIME, Eli5, QLattice, and Anchor, achieving an accuracy of 89%. Goenka et al. [12] employed a ConvNet for Alzheimer's detection, reaching 97% accuracy, while Palkar et al. [13] applied XGBoost with XAI integration for glioma prognosis, obtaining 88% accuracy. Similarly, Vivek Khanna et al. [14] employed a stacking ensemble model for the detection of gestational diabetes, achieving 96% accuracy while utilizing multiple XAI techniques to enhance interpretability.

TABLE 7 | Comparison of XAI-based feature importance and clinical expert insights for MHR factors.

Priority	XAI decision			Clinical expert decision	
	SHAP	LIME	FORCE	Expert—1	Expert—2
First	Blood sugar	Body temperature	Blood sugar	Age, both systolic and diastolic Blood pressure, and Blood Sugar are, in general, related to risk factors in pregnancy.	Any blood pressure
Second	SystolicBP	SystolicBP	Body temperature		Blood sugar
Third	Age	Age	SystolicBP		—
Fourth	Body temperature	Blood sugar	Age		—
Fifth	DiastolicBP	Heart rate	DiastolicBP		—
Sixth	Heart rate	DiastolicBP	—		—

TABLE 8 | Comparison of this research with existing methods.

Methods	Technique	Accuracy	Cross validation	XAI
Mutlu et al. [5]	DT	89.16%	No	No
Raza et al. [6]	SVM + DT-BiLTCN	98%	No	No
Mondal et al. [7]	RF	93.14%	No	No
Togunwa et al. [8]	SVM	95%	No	No
Nirmala and Kambli [9]	RF	96%	No	No
Ahmed and Kashem [10]	DT	97%	No	No
Chadaga et al. [11]	Stacking ensemble	89%	Yes (5 fold)	Yes
Goenka et al. [12]	ConvNet	97%	No	No
Palkar et al. [13]	XGBoost	88%	Yes (5 fold)	Yes
Vivek Khanna et al. [14]	Stacking ensemble	96%	Yes (5 fold)	Yes
Our method	Bagging ensemble model + PCA	99.51%	Yes (10 fold)	Yes

In contrast, our study integrates XAI techniques (SHAP and LIME) and feature optimization methods, including LDA, PCA, and RFE. Moreover, we compare the decisions of our XAI model with those of clinical experts and doctors, ensuring its alignment with expert judgment. Additionally, our research provides real-time insights into clinical healthcare settings, addressing practical challenges and offering training for doctors to effectively utilize the model. Our approach not only surpasses the accuracy of previous studies but also adds extra trustworthiness through 10-fold cross-validation, standard deviation, and a 95% confidence interval. This combination of technical rigor and clinical applicability makes our study more effective and impactful, offering significant advantages for MHR prediction.

Figure 14 depicts a real-time overview of the MHR AI System. It illustrates the process of a pregnant patient visiting a doctor, where key health parameters, such as blood pressure, glucose levels, temperature, and vital signs, are measured. These parameters are fed into the AI system, which analyzes the optimal models designed for MHR assessment. The system integrates the most relevant features to predict whether the patient is at low, medium, or high risk for health complications. The doctor can use this AI-generated prediction to make informed decisions on patient care. Additionally, the AI's decision-making process can be compared with the doctor's assessment using XAI, allowing medical professionals and non-experts to understand and evaluate the risk management decisions. This enables more

informed decision-making and promotes transparency in the AI's conclusions. However, urban healthcare facilities, equipped with modern infrastructure, may face fewer challenges, as most already have the necessary computers to support such systems. In contrast, rural areas may struggle with electricity availability and the need for properly configured computers to implement the system effectively. To address these concerns, we outline the steps for seamless integration into traditional healthcare systems, as visualized in Figure 11. Fortunately, no significant training is required for medical professionals to use the system; an AI expert can easily demonstrate how to input patient data into the computer. Additionally, privacy concerns are a key focus of our ongoing work. We are actively working on implementing differential privacy techniques to protect patient information, ensuring data security while maintaining the system's efficacy. In future work, we aim to refine these privacy solutions and further streamline integration, particularly in resource-limited settings, to ensure the system benefits all patients, regardless of location.

Table 9 presents the performance metrics of our best model, Bagging combined with PCA, which we identify as the most effective approach after evaluating several ML algorithms for MHR prediction. To ensure external validation, we apply SMOTE to balance the dataset and enhance its robustness. SMOTE is an advanced data augmentation technique that generates synthetic samples for minority classes by interpolating between existing samples,

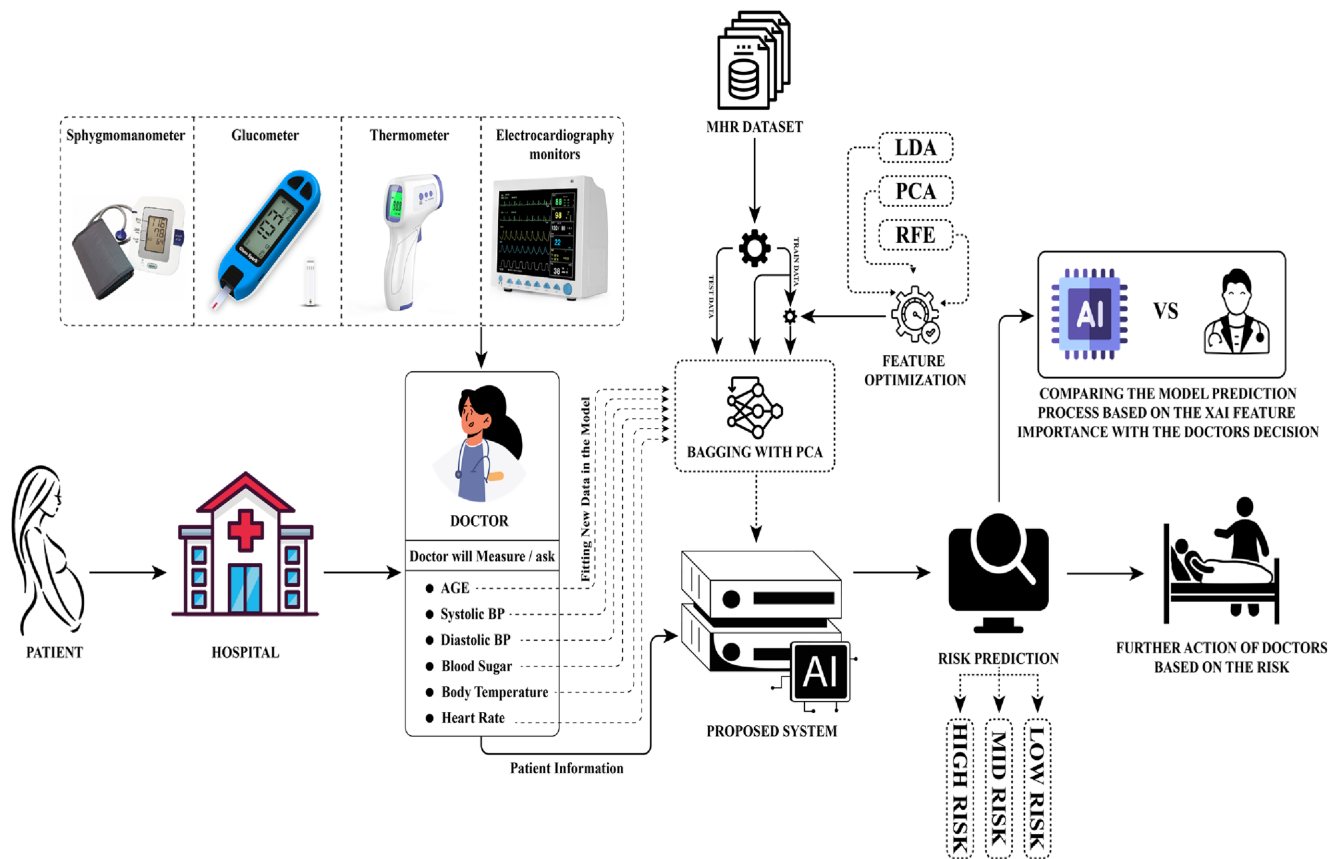


FIGURE 14 | Real-time risk assessment and XAI for improved decision-making.

TABLE 9 | Performance metrics of the Bagging + PCA model for MHR prediction with a generative dataset.

Fold/metric	Accuracy	Precision	Recall	F1-score
1	90.91	90.91	90.91	90.77
2	91.18	92.38	91.18	90.97
3	91.18	92.38	91.18	90.97
4	93.94	94.87	93.94	93.94
5	94.12	95.19	94.12	94.29
6	94.12	95.02	94.12	94.00
7	97.06	97.30	97.06	97.07
8	100.00	100.00	100.00	100.00
9	100.00	100.00	100.00	100.00
10	100.00	100.00	100.00	100.00
Mean	95.25	95.80	95.25	95.20
SD	3.76	3.40	3.76	3.83
CI	(92.92, 97.58)	(93.69, 97.92)	(92.92, 97.58)	(92.83, 97.57)

improving the model’s ability to learn from underrepresented data and thereby increasing predictive accuracy [54]. Initially, our dataset contains 1015 samples, with 406 linked to low risk, 336 to mid risk, and 272 to high risk. We upscale each class by 500 to achieve a balanced dataset, resulting in 1500 samples. After this augmentation, we remove the original 1015 samples and retain the newly generated 485 samples, creating a revised dataset. We then fit the primary dataset to the training set of our best model,

Bagging + PCA, and use the new 485-sample dataset for testing and outcome prediction.

We use feature optimization despite our dataset comprising only six features because it effectively reduces overfitting. Additionally, as our dataset is imbalanced, feature optimization proves beneficial in addressing this imbalance. It also manages outliers, enabling better training of ML models [55, 56].

TABLE 10 | Threats to validity and validation strategies used in this study.

Validity type	Potential threat	How we addressed it (Validation performed)	Remaining limitations
Internal validity	Overfitting or training bias	10-fold cross-validation, 80/20 train-test split, grid search tuning, L1/L2 regularization	The dataset size is relatively small
External validity	Limited generalization to other populations or hospitals	External validation using the SMOTE-balanced dataset; comparison with other studies in the literature	Single-source dataset; no multi-center clinical validation yet
Construct validity	Incorrect definition of features or risk labels	Risk levels aligned with clinical guidelines; clinical expert consultation; feature optimization (PCA, LDA, RFE)	Some additional maternal factors (e.g., lifestyle, medical history) were not included.
Conclusion validity	Misinterpretation of model performance or unstable results	Multiple evaluation metrics (Accuracy, precision, recall, F1, ROC), standard deviation, and 95% confidence interval across folds confirm stability and reliability	No formal hypothesis-based significance test (e.g., <i>t</i> -test/ANOVA), but CI and SD already provide strong statistical reliability

Table 10 summarizes the key threats to the validity of this study and outlines the strategies implemented to mitigate them. To ensure internal validity, the model was trained using 10-fold cross-validation, an 80/20 train-test split, and grid search-based hyperparameter tuning with L1/L2 regularization. These techniques reduce the risk of overfitting and enhance the model's generalizability. However, we acknowledge that the dataset size is relatively small, which may still introduce minor bias. In terms of external validity, although the proposed model was validated using a SMOTE-balanced dataset and compared against existing studies in the literature, the dataset originates from a single source. Therefore, future multi-center or multi-population validation is necessary for broader clinical applicability. Construct validity was strengthened by aligning risk levels with clinical guidelines, selecting medically relevant features, and consulting clinical experts to confirm the appropriateness of feature definitions and model interpretations. Nevertheless, additional maternal health factors such as lifestyle, prior medical history, or genetic data were not included and could improve representativeness in future work. Finally, conclusion validity was reinforced using multiple evaluation metrics (Accuracy, Precision, Recall, F1, and ROC) and by reporting Standard Deviation and 95% Confidence Interval across folds, confirming strong model stability and reliability. While no formal hypothesis-based statistical test (e.g., *t*-test or ANOVA) was conducted, the SD and CI values already provide robust statistical evidence supporting the consistency of the model's performance. Overall, the strategies presented in Table 10 demonstrate that this study carefully addressed major threats to validity, ensuring the trustworthiness and rigor of the proposed maternal health risk prediction framework.

5 | Conclusion

This study presents an automated system that uses ML and XAI to address MHR, a critical global issue. It highlights the importance of timely interventions during the prenatal and postpartum periods. The study enhances MHR detection by employing

datasets and advanced preprocessing techniques. Ten ML models are evaluated for effectiveness using various feature optimization methods, with bagging combined with PCA emerging as the best model for automatic MHR detection. Performance evaluation metrics confirm the research's accuracy, and the use of XAI tools, such as SHAP and LIME plots, enhances the interpretability of the decision-making process. This research underscores the significance of ML and XAI in tackling healthcare challenges, aiming to reduce maternal mortality worldwide. However, a small dataset and limited features reduce the model's generalizability, potentially overlooking important maternal health risk factors in this research. Future efforts will focus on integrating more advanced XAI methods and applying this approach in real-world clinical settings.

Author Contributions

Mohammad Mamun: conceptualization, investigation, writing – original draft, methodology, visualization, writing – review and editing, formal analysis, data curation. **Safiul Haque Chowdhury:** conceptualization, investigation, writing – original draft, writing – review and editing, methodology, validation, visualization, software, formal analysis, investigation, resources. **Md. Omar Faruq:** writing – review and editing, visualization, resources. **Mir Mohammad Azad:** writing – review and editing, validation, resources. **Bishakha Rani Biswas:** writing – review and editing. **Mohammed Ibrahim Hussain:** writing – review and editing, supervision. **Muhammad Minoar Hossain:** supervision, writing – review and editing, validation, investigation, conceptualization, methodology.

Acknowledgments

We want to express our heartfelt thanks to Sayma Siddiqua (Expert 1), who works at Leicester Royal Infirmary, Leicester, England, and is a former doctor at Bangabandhu Sheikh Mujib Medical University, Dhaka, Bangladesh and to Rahat Fahmida Hridi (Expert 2), an Honorary Medical Officer at MU-VA at Dhaka Medical College Hospital, who also works at the National Institute of Mental Health and the Ministry of Health and Family Welfare as an Assistant Surgeon. We are deeply grateful for their medical expertise, which enabled us to compare our XAI-based

decision-making processes with those made by experienced doctors. Their contributions were invaluable to the success of this research.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. M. C. Hogan, K. J. Foreman, M. Naghavi, et al., “Maternal Mortality for 181 Countries, 1980–2008: A Systematic Analysis of Progress Towards Millennium Development Goal 5,” *Lancet* 375, no. 9726 (2010): 1609–1623.
2. M. A. Caudill-Slosberg, “The Human Journey,” Health and Education in the Modern World, 2019, <https://humanjourney.us/health/maternal-and-child-health/>.
3. L. Say, D. Chou, A. Gemmill, et al., “Global Causes of Maternal Death: A WHO Systematic Analysis,” *Lancet Global Health* 2, no. 6 (2014): e323–e333.
4. A. Esteva, A. Robicquet, B. Ramsundar, et al., “A Guide to Deep Learning in Healthcare,” *Nature Medicine* 25, no. 1 (2019): 24–29.
5. H. B. Mutlu, F. Durmaz, N. Yücel, E. Cengil, and M. Yildirim, “Prediction of Maternal Health Risk With Traditional Machine Learning Methods,” *Naturens* 4, no. 1 (2023): 16–23.
6. A. Raza, H. U. R. Siddiqui, K. Munir, M. Almutairi, F. Rustam, and I. Ashraf, “Ensemble Learning-Based Feature Engineering to Analyze Maternal Health During Pregnancy and Health Risk Prediction,” *PLoS One* 17, no. 11 (2022): e0276525.
7. S. Mondal, A. Nag, A. K. Barman, and M. Karmakar, “Machine Learning-Based Maternal Health Risk Prediction Model for IoMT Framework,” *International Journal of Experimental Research and Review* 32 (2023): 145–159.
8. T. O. Togunwa, A. O. Babatunde, and K. U. R. Abdullah, “Deep Hybrid Model for Maternal Health Risk Classification in Pregnancy: Synergy of ANN and Random Forest,” *Frontiers in Artificial Intelligence* 6 (2023): 1213436.
9. Nirmala and R. S. Kambli, “Model for Predicting Risk Levels in Maternal Healthcare,” *International Journal of Advance Research and Innovative Ideas in Education* 8, no. 6 (2022): 1633–1637.
10. M. Ahmed and M. A. Kashem, “IoT-Based Risk Level Prediction Model for Maternal Health Care in the Context of Bangladesh,” in *2020 2nd International Conference on Sustainable Technologies for Industry 4.0 (STI)* (IEEE, 2020), 1–6.
11. K. Chadaga, S. Prabhu, V. Bhat, N. Sampathila, S. Umakanth, and R. Chadaga, “Artificial Intelligence for Diagnosis of Mild–Moderate COVID-19 Using Haematological Markers,” *Annals of Medicine* 55, no. 1 (2023): 2233541.
12. N. Goenka, A. K. Sharma, S. Tiwari, et al., “A Regularized Volumetric ConvNet Based Alzheimer Detection Using T1-Weighted MRI Images,” *Cogent Engineering* 11, no. 1 (2024): 2314872.
13. A. Palkar, C. C. Dias, K. Chadaga, and N. Sampathila, “Empowering Glioma Prognosis With Transparent Machine Learning and Interpretative Insights Using Explainable AI,” *IEEE Access* 12 (2024): 31697–31718.
14. V. Vivek Khanna, K. Chadaga, N. Sampathila, et al., “Explainable Artificial Intelligence-Driven Gestational Diabetes Mellitus Prediction Using Clinical and Laboratory Markers,” *Cogent Engineering* 11, no. 1 (2024): 2330266.
15. J. Arvidsson, “2023,” <https://www.kaggle.com/datasets/joebeach/capital/maternal-health-risk>.
16. R. Sivanesan and K. D. R. Dhivya, “A Review on Diabetes Mellitus Diagnoses Using Classification on Pima Indian Diabetes Data Set,” *International Journal of Advance Research in Computer Science and Management Studies* 5, no. 1 (2017).
17. G. F. C. Campos, S. M. Mastelini, G. J. Aguiar, R. G. Mantovani, L. F. D. Melo, and S. Barbon, “Machine Learning Hyperparameter Selection for Contrast Limited Adaptive Histogram Equalization,” *EURASIP Journal on Image and Video Processing* 2019 (2019): 1–18.
18. R. Tanious and R. Manolov, “Violin Plots as Visual Tools in the Meta-Analysis of Single-Case Experimental Designs,” *Methodology* 18, no. 3 (2022): 221–238.
19. A. Pryke, S. Mostaghim, and A. Nazemi, “Heatmap Visualization of Population Based Multi Objective Algorithms,” in *Evolutionary Multi-Criterion Optimization: 4th International Conference, EMO 2007, Matsushima, Japan, March 5–8, 2007. Proceedings 4* (361–375). (Springer Berlin Heidelberg, 2007).
20. G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning*, vol. 112 (Springer, 2013), 18.
21. C. M. Bishop and N. M. Nasrabadi, *Pattern Recognition and Machine Learning*, vol. 4 (springer, 2006), 738.
22. Y. Tian and Y. Zhang, “A Comprehensive Survey on Regularization Strategies in Machine Learning,” *Information Fusion* 80 (2022): 146–166.
23. T. Chen, T. He, M. Benesty, et al., “Xgboost: Extreme Gradient Boosting,” *R Package Version 0.4-2* 1, no. 4 (2015): 1–4.
24. M. Belgiu and L. Drăguț, “Random Forest in Remote Sensing: A Review of Applications and Future Directions,” *ISPRS Journal of Photogrammetry and Remote Sensing* 114 (2016): 24–31.
25. L. Breiman, “Bagging Predictors,” *Machine Learning* 24 (1996): 123–140.
26. R. E. Schapire, “A Brief Introduction to Boosting,” *International Joint Conferences on Artificial Intelligence* 112, no. 999 (1999): 1401–1406.
27. B. Pavlyshenko, “Using Stacking Approaches for Machine Learning Models,” in *2018 IEEE Second International Conference on Data Stream Mining & Processing (DSMP)* (IEEE, 2018), 255–258.
28. D. Burka, C. Puppe, L. Szepesváry, and A. Tasnádi, “Voting: A Machine Learning Approach,” *European Journal of Operational Research* 299, no. 3 (2022): 1003–1017.
29. A. Natekin and A. Knoll, “Gradient Boosting Machines, a Tutorial,” *Frontiers in Neurorobotics* 7 (2013): 21.
30. J. Fan, X. Ma, L. Wu, F. Zhang, X. Yu, and W. Zeng, “Light Gradient Boosting Machine: An Efficient Soft Computing Model for Estimating Daily Reference Evapotranspiration With Local and External Meteorological Data,” *Agricultural Water Management* 225 (2019): 105758.
31. T. Zhang, Y. Huang, H. Liao, and Y. Liang, “A Hybrid Electric Vehicle Load Classification and Forecasting Approach Based on GBDT Algorithm and Temporal Convolutional Network,” *Applied Energy* 351 (2023): 121768.
32. G. Louppe and P. Geurts, “Ensembles on Random Patches,” in *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2012, Bristol, UK, September 24–28, 2012. Proceedings, Part I* 23 (346–361). (Springer Berlin Heidelberg, 2012).
33. C. Cortes, M. Mohri, and D. Storcheus, “Regularized Gradient Boosting,” *Advances in Neural Information Processing Systems* 32 (2019): 1–10.

34. J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the Random Forest Framework for Classification of Hyperspectral Data," *IEEE Transactions on Geoscience and Remote Sensing* 43, no. 3 (2005): 492–501.
35. P. Bühlmann and B. Yu, "Analyzing Bagging," *Annals of Statistics* 30, no. 4 (2002): 927–961.
36. R. E. Schapire, "A Brief Introduction to Boosting," *Ijcai* 99, no. 999 (1999): 1401–1406.
37. Z. Malá, L. Křivánková, P. Gebauer, and P. Boček, "Contemporary Sample Stacking in CE: A Sophisticated Tool Based on Simple Principles," *Electrophoresis* 28, no. 1–2 (2007): 243–253.
38. B. Parhami, "Voting Algorithms," *IEEE Transactions on Reliability* 43, no. 4 (1994): 617–629.
39. V. K. Ayyadevara, "Gradient Boosting Machine," in *Pro Machine Learning Algorithms: A Hands-On Approach to Implementing Algorithms in Python and R* (Apress, 2018), 117–134.
40. R. M. Aziz, M. F. Baluch, S. Patel, and A. H. Ganie, "LGBM: A Machine Learning Approach for Ethereum Fraud Detection," *International Journal of Information Technology* 14, no. 7 (2022): 3321–3331.
41. W. Liang, S. Luo, G. Zhao, and H. Wu, "Predicting Hard Rock Pillar Stability Using GBDT, XGBoost, and LightGBM Algorithms," *Mathematics* 8, no. 5 (2020): 765.
42. H. M. Gomes, J. Read, and A. Bifet, "Streaming Random Patches for Evolving Data Stream Classification," in *2019 IEEE International Conference on Data Mining (ICDM)* (IEEE, 2019), 240–249.
43. C. Room, "Confusion Matrix," *Machine Learning* 6 (2019): 27.
44. S. C. Asogwa, F. C. Obodoeze, E. A. Etuk, and N. E. Ugwuja, "Smart City PM2.5 Air Pollution Modeling Techniques: Train-Test Data Split Versus K-Fold Cross Validation Techniques," *Journal of Inventive Engineering and Technology (JIET)* 3, no. 1 (2023): 30–44.
45. H. T. T. Nguyen, H. Q. Cao, K. V. T. Nguyen, and N. D. K. Pham, "Evaluation of Explainable Artificial Intelligence: Shap, Lime, and Cam," *Proceedings of the FPT AI Conference* (2021): 1–6.
46. S. H. Chowdhury, M. Mamun, T. A. Shaikat, M. I. Hussain, S. Iqbal, and M. M. Hossain, "An Ensemble Approach for Artificial Neural Network-Based Liver Disease Identification From Optimal Features Through Hybrid Modeling Integrated With Advanced Explainable AI," *Medinfo* 2, no. 2 (2025): 107–119.
47. D. G. Altman and J. M. Bland, "Standard Deviations and Standard Errors," *BMJ (Clinical Research Ed.)* 331, no. 7521 (2005): 903.
48. G. Cumming, "Inference by Eye: Reading the Overlap of Independent Confidence Intervals," *Statistics in Medicine* 28, no. 2 (2009): 205–220.
49. R. Kohavi, *A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection* (Morgan Kaufman Publishing, 1995).
50. M. Abdulla-Hil-Kafi, M. A. Sheakh, M. A. Oishe, M. M. Islam, M. M. R. Foysal, and T. Sarwar, "Classification of Maternal Health Risk Factors Using Machine Learning Approach," in *2024 International Conference on Social and Sustainable Innovations in Technology and Engineering (SASI-ITE)* (IEEE, 2024), 1–6.
51. D. Thakkar, V. C. Gandhi, and D. Trivedi, "Forecasting Maternal Women's Health Risks Using Random Forest Classifier," in *2024 International Conference on Inventive Computation Technologies (ICICT)* (IEEE, 2024), 961–965.
52. D. Bajaj, R. Kumari, and P. Bansal, "Risk Level Prediction for Maternal Health Using Machine Learning Algorithms," in *2023 International Conference on Communication, Security and Artificial Intelligence (ICCSAI)* (IEEE, 2023), 405–409.
53. B. U. Maheswari, A. Dixit, and A. K. Karn, "Machine Learning Algorithm for Maternal Health Risk Classification With SMOTE and Explainable AI," in *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)* (IEEE, 2024), 1–6.
54. N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-Sampling Technique," *Journal of Artificial Intelligence Research* 16 (2002): 321–357.
55. A. K. Kim and H. Chung, "The Effect of Rebalancing on LDA in Imbalanced Classification," *Stat* 10, no. 1 (2021): e384.
56. P. Filzmoser, K. Hron, and C. Reimann, "Principal Component Analysis for Compositional Data With Outliers," *Environmetrics* 20, no. 6 (2009): 621–632.